01.02 · THEORY + PAST-PAPER INDEX

Electromagnetic Theory & Quantum Physics

Light, radiation and the quantum packet

START HERE

Astronomy is built from light: nearly everything we know about a distant object arrives as electromagnetic radiation. This lecture starts with waves and photons, develops flux and luminosity, and derives the blackbody laws used to infer temperatures and energy output.

LEARNING GOALS

✓	Move between wavelength, frequency and photon energy.
✓	Distinguish luminosity, flux and intensity at an observing distance.
✓	Use Planck, Wien and Stefan–Boltzmann laws in their proper roles.
✓	Explain the quantum meaning of radiation and the photoelectric threshold.

1 · Electromagnetic waves and the spectrum

An electromagnetic wave is a changing electric field and magnetic field that carries energy through space. In vacuum every frequency travels at the speed of light $c = 2.998 \times 10^{8}$ m s^{-1}. Wavelength λ is the distance between successive wave crests; frequency ν is the number of cycles passing a point per second.

$$c = \lambda\nu$$

The electromagnetic spectrum is ordered by wavelength or, equivalently, by frequency: radio, microwave, infrared, visible, ultraviolet, X-ray and gamma ray. Shorter wavelength means higher frequency and greater photon energy. The boundaries between bands are conventional, not physical discontinuities.

ASTRONOMICAL MEANING

Different wavelengths reveal different physical conditions. Cool dust is prominent in the infrared; neutral hydrogen emits a famous radio line near 21 cm; hot gas can shine in X-rays. Earth's atmosphere transmits some windows well and blocks much of the ultraviolet, X-ray and gamma-ray sky.

A measured wavelength is often quoted in nanometres (1 nm = 10^{-9} m), micrometres (1 μm = 10^{-6} m) or centimetres. Convert to metres before using SI constants, unless the formula is deliberately written in matching units.

2 · Photons: the quantum of electromagnetic energy

Quantum physics assigns electromagnetic radiation energy in packets called photons. A photon has no rest mass, but it carries energy and momentum. The energy depends on frequency, not on how many photons are present.

$$E_\gamma = h\nu = hc/\lambda \cdot p_\gamma = h/\lambda$$

Planck's constant is $h = 6.626 \times 10^{-34}$ J s. A useful conversion is $hc \approx 1240$ eV nm, so a photon of wavelength λ in nanometres has energy E in electronvolts given by $E(\text{eV}) \approx 1240/\lambda(\text{nm})$. One electronvolt is 1.602×10^{-19} joule.

INTENSITY VERSUS PHOTON ENERGY

Increasing the intensity of a monochromatic beam usually increases the number of photons arriving per second; it does not change the energy of each photon. Changing the wavelength changes each photon's energy. Keep these ideas separate when reasoning about detectors and the photoelectric effect.

In the photoelectric effect, one photon gives energy $h\nu$ to one electron. If the photon energy exceeds a material's work function φ, the electron can escape; the maximum kinetic energy is $K_{max} = h\nu - \varphi$. Below the threshold frequency $\nu_0 = \varphi/h$, increasing intensity alone cannot eject electrons in the ideal one-photon model.

3 · Luminosity and flux: source power versus received power

Luminosity L is the total radiant energy emitted by a source per second, measured in watts. Flux F is energy arriving per second per unit area, also measured in W m^{-2}. For an isotropic source, its power spreads over larger and larger spherical surfaces as the distance grows.

$$F(r) = L / (4\pi r^2) \cdot L = 4\pi r^2 F$$

This inverse-square law does not mean the star's luminosity decreases with distance; it means the same emitted power is distributed over an area proportional to r^2. If distance doubles, flux falls by a factor of four. Interstellar absorption or beaming can change the measured flux from the simple isotropic prediction.

OBSERVED BAND AND BOLOMETRIC OUTPUT

A detector usually measures only a wavelength interval. Bolometric flux is the total flux integrated over all wavelengths. A source can be bright in one band and faint in another, so state whether a flux or luminosity is monochromatic, band-limited or bolometric.

For a spherical star, surface area is $4\pi R^2$. If the emitted surface flux is $F_{surface}$, its luminosity is $L = 4\pi R^2 F_{surface}$. This connects measurements at Earth to the physical size and temperature of the emitting surface.

4 · Thermal radiation and the blackbody model

A blackbody is an ideal object that absorbs all incident radiation and emits a spectrum determined only by its temperature. Real stars are not perfect blackbodies, but their broad continua are often close enough for a first estimate. The Planck distribution describes how energy is spread across wavelength.

$$B_\lambda(T) = [2hc^2/\lambda^5] \cdot 1/[\exp(hc/(\lambda k_B T)) - 1]$$

In this expression B_λ is spectral radiance per unit wavelength and solid angle, and $k_B = 1.381 \times 10^{-23}$ J K^{-1}. The full formula need not be evaluated in every problem. Its two most useful consequences are Wien's displacement law and the Stefan–Boltzmann law.

$$\lambda_{max} T = b \cdot F_{surface} = \sigma T^4 \cdot L = 4\pi R^2 \sigma T^4$$

Wien's constant is $b = 2.898 \times 10^{-3}$ m K. Hotter blackbodies peak at shorter wavelengths. The Stefan–Boltzmann constant is $\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}; total emitted surface flux rises very steeply with temperature. If radius is fixed and temperature doubles, luminosity increases by $2^4 = 16$.

CHOOSING THE RIGHT LAW

Use Wien's law to connect temperature with the peak wavelength. Use Stefan–Boltzmann to relate temperature to total emitted flux or luminosity. Use Planck's law when a question asks for a spectrum at a specific wavelength. These are related laws, but they answer different questions.

5 · Radiation pressure and energy transport

Radiation carries momentum as well as energy. When a beam of flux F is completely absorbed by a surface, its radiation pressure is F/c; for ideal reflection the momentum change doubles and the pressure is 2F/c. For most planets and ordinary stars, gravity dominates the motion of large bodies, but radiation pressure can matter for dust grains and very luminous stars.

$$P_{rad} = F/c \text{ (absorption)} \cdot P_{rad} = 2F/c \text{ (ideal reflection)}$$

Radiation can move energy through empty space, while conduction requires material contact and convection transports energy through bulk fluid motion. In a star, photons can be absorbed and re-emitted many times before escaping; the surface spectrum therefore reflects the outer layers, not an instantaneous view of the core.

A USEFUL THREE-STEP MEASUREMENT

(1) Infer a colour or peak wavelength. (2) Estimate temperature with Wien's law. (3) Combine the temperature with a radius or flux measurement through Stefan–Boltzmann and the inverse-square law. Check that every quantity refers to the same wavelength coverage and distance.

PAST IOAA PROBLEM REFERENCES

The following questions are from official past IOAA problem sets.

Use each year, round and question label to locate the original problem in its official paper.

01	IOAA 2007 · TH 1.8
02	IOAA 2007 · TH 1.9
03	IOAA 2007 · TH 1.10
04	IOAA 2007 · TH 1.11
05	IOAA 2007 · TH 1.15
06	IOAA 2008 · TH L2
07	IOAA 2009 · TH P2
08	IOAA 2010 · TH S1
09	IOAA 2011 · TH S8
10	IOAA 2011 · TH S9
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15	IOAA 2014 · TH P8
16	IOAA 2014 · TH P9
17	IOAA 2015 · TH S6
18	IOAA 2015 · TH S8
19	IOAA 2017 · TH T4
20	IOAA 2017 · TH T6

21 IOAA 2018 · TH T6
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24 IOAA 2022 · TH 11
25 IOAA 2023 · TH Q2
26 IOAA 2025 · TH T04